

Problem-Solution Fit

Date	30 June 2026
Team ID	*****
Project Name	Smart Lender – AI-Powered Loan Eligibility Prediction System
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

The Problem–Solution Fit Canvas highlights the challenges in loan approval and the proposed AI-based solution. It helps improve accuracy, speed, and decision-making

1. CUSTOMER SEGMENT(S) [CS] - Bank loan applicants - Credit officers - Financial institutions - Banks and NBFCs - Small business loan applicants	6. CUSTOMER LIMITATIONS (EG. BUDGET, DEVICES) [CL] Loan applicants and bank officials face several challenges during the loan approval process, including incomplete information, lengthy verification procedures, and limited awareness of eligibility criteria. These limitations reduce efficiency and increase processing time.	5. AVAILABLE SOLUTIONS PROS & CONS [AS] Existing loan approval systems mainly rely on manual verification and predefined rules to evaluate applicants. Although these methods are reliable for document verification, they are time-consuming, require significant human effort, and may lead to delays or inconsistent decisions.
2. PROBLEMS / PAINS + ITS FREQUENCY [PR] - Manual loan verification consumes significant time. - Loan approval decisions may be inconsistent. - Large numbers of applications increase processing delay	9. PROBLEM ROOT / CAUSE [RC] No standardized, data-driven way to assess creditworthiness; decisions depend on individual officer judgment, leading to inconsistent and error-prone approvals.	7. BEHAVIOR + ITS INTENSITY [BE] Loan applicants expect a quick, transparent, and accurate approval process, while bank officials handle a large number of applications every day. This creates a strong demand for an automated system.
3. TRIGGERS TO ACT [TR] - Need for faster loan approval. - Growing number of loan applications. - Requirement for accurate decision-making. - Reducing manual effort and paperwork.	10. YOUR SOLUTION [SL] Smart Lender: a Flask-based web app that uses a trained XGBoost model to instantly predict loan approval likelihood from applicant income, credit history, and asset data, cutting review time from days to seconds.	8. CHANNELS OF BEHAVIOR [CH] ONLINE Bank intranet portal, internal loan-processing dashboard, email notifications for flagged high-risk applications.
4. EMOTIONS BEFORE / AFTER [EM] Before: Worried about loan approval, Frustrated with long processing time. After: Confident in fair evaluation.		OFFLINE In-branch consultations, printed application review meetings, regulatory/compliance audits.

Stisfies with quick prediction,Trust the AI-based decision process			
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